XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

COM105 Engineering Physics (I)

Lecture 4: Dynamics - Newton's Laws of Motion

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In this chapter, you'll learn…

- what the concept of force means in physics, why forces are vectors, and the significance of the net force on an object.
- what happens when the net force on an object is zero, and the significance of inertial frames of reference.
- how the acceleration of an object is determined by the net force on the object and the object's mass.
- the difference between the mass of an object and its weight.
- how the forces that two objects exert on each other are related.

Introduction to Dynamics

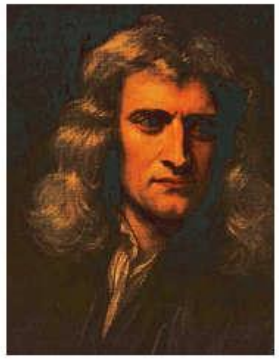
- So far we have learned how to use kinematics to describe motion in one, two, or three dimensions.
- But what causes bodies to move the way that they do?
  - Why does a dropped feather fall more slowly than a dropped baseball?
  - Why do you feel pushed backward in a car that accelerates forward?
- The answers to such questions take us into the subject of dynamics, the relationship of motion to the forces that cause it.

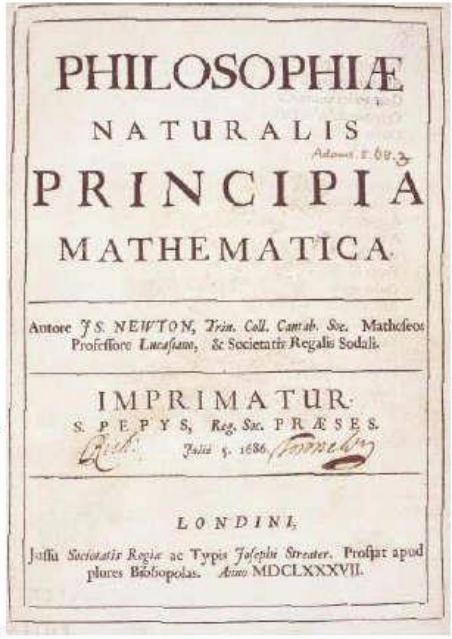


Sir Isaac Newton (1642–1727)

- A great British Physicist
- The principles of dynamics were clearly stated for the first time by Sir Isaac Newton; today we call them Newton's laws of motion.
- Newton did not derive these laws, but rather deduced them from a multitude of experiments performed by other scientists.

Philosophiae Naturalis Principia Mathematica, Issac Newton, 1687




PHILOSOPHIÆ
NATURALIS
PRINCIPIA
MATHEMATICA
Autore J S. NEWTON, Trin. Coll. Cantab. Soc. Matheseos Professore Lucasiano, & Societatis Regalis Sodali.
IMPRIMATUR
S. PEPYS, Reg. Soc. PRÆSES.
Julii 5. 1686.
LONDINI,
Jussu Societatis Regiæ ac Typis Josephi Streater. Prostat apud plures Bibliopolas. Anno MDCLXXXVII.

You may find pdf versions of the Principia in the Internet, eg, http://www.archive.org/details/newtonspmathema00newtrich

Newton's Laws of Motion

- Newton's Laws of Motion
  - Newton's First Law
  - Newton's Second Law
  - Newton's Third Law
  - ➢ Describe the relationship between the motion of a body and mass and **force** acting on it.
- Newton's laws of motion are the foundation of Newtonian Mechanics /Classical Mechanics.
  - ➢ only dealing with objects that are:
    - i. Larger than atomic dimensions ($>>10^{-10}$ m) but smaller than 10^{10} m, i.e., dark matter in galaxies
    - ii. Move at speeds that are much lower than speed of light (c)

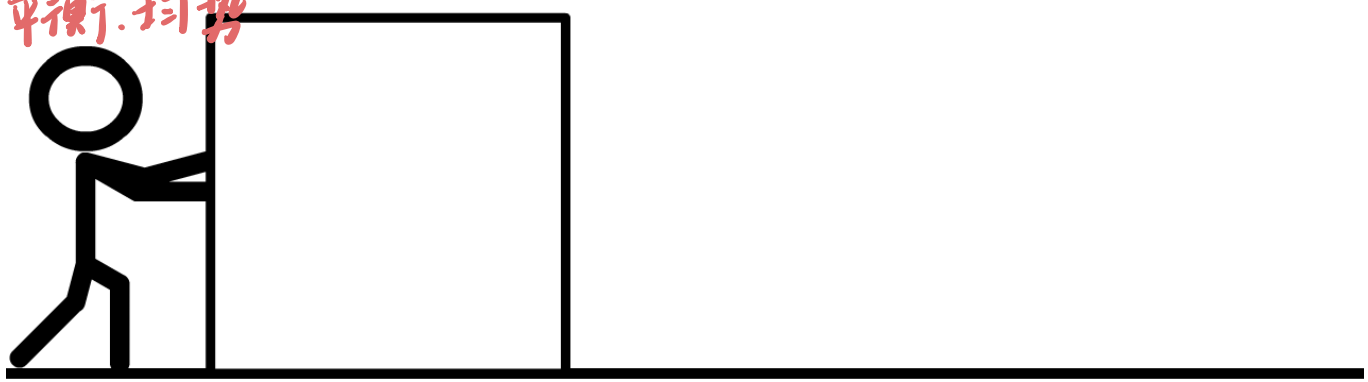
Force

- Force represents the interaction of an object with its environment.
- Force is associated with the result of “muscular” activity and/or some change in the motion of an object.
- A net non-zero force on an object causes it to accelerate.
- When an object is at rest or uniform velocity, it is said to be in equilibrium.

静态

匀速

平衡.均势

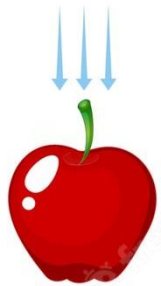


Force

- Examples



FRICTION FORCE



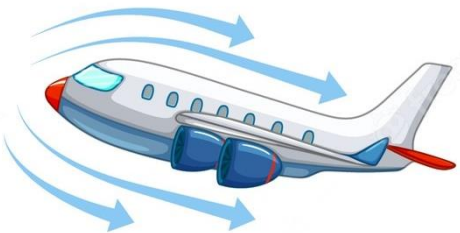
GRAVITY FORCE



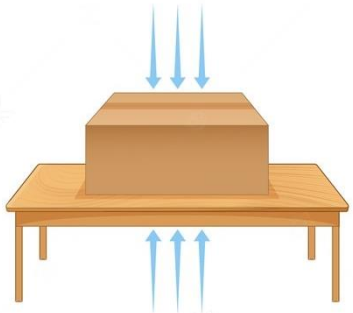
APPLIED FORCE



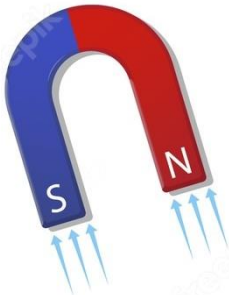
SPRING FORCE



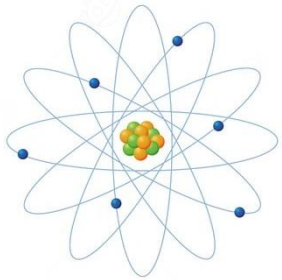
DRAG FORCE



NORMAL FORCE



MAGNETIC FORCE



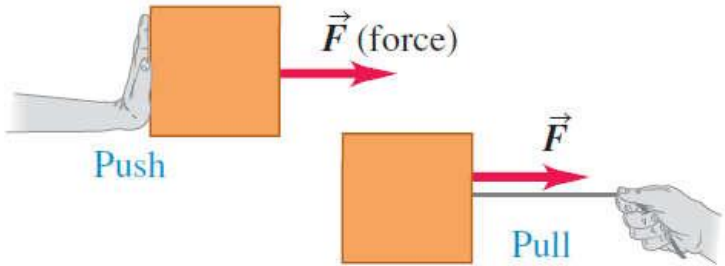
ELECTRIC FORCE

Force

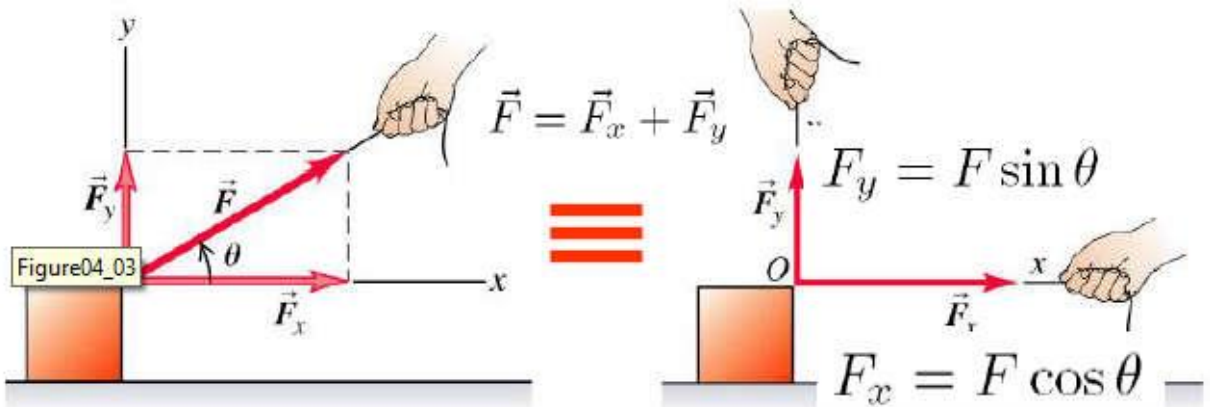
- Contact forces
  - Represent the result of physical contact between two objects (i.e., pulling, compressing a spring, etc.)
  - Only present during the “contact”. Does not remain in the body when the “contact” is over.
- Field forces
  - Do not involve physical contact between two objects but act through empty space (i.e., gravitational, electrical forces, etc.)
  - At the atomic level, all the forces classified as contact forces are actually electrical (field) forces.

Force

- In everyday language, force can be push or pull.

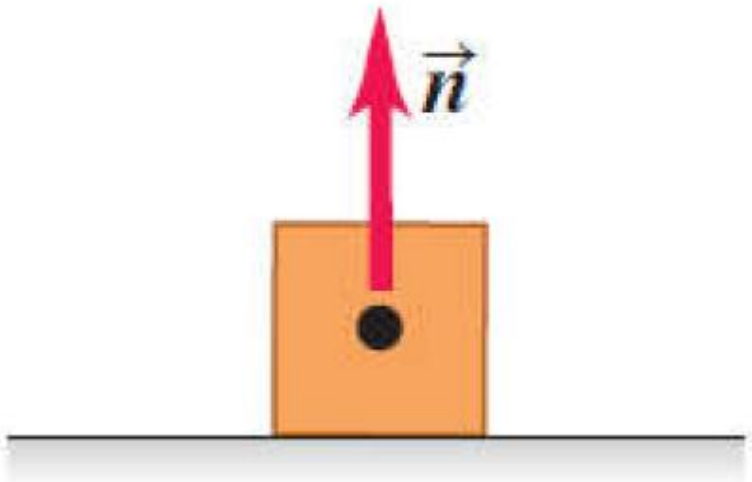


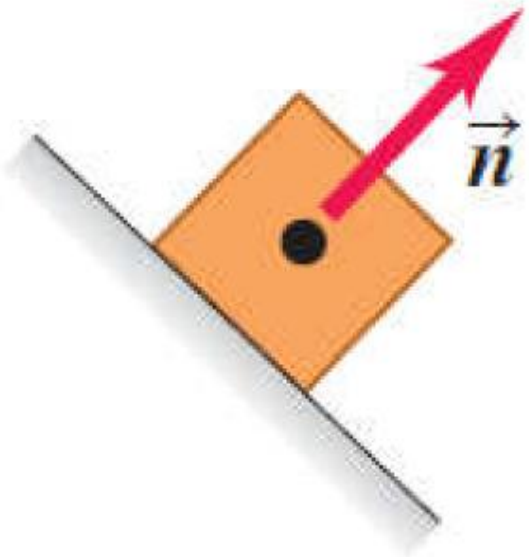
- Force is a vector with magnitude and direction. Unit: N (Newton).



Normal Force 正交力、法向力

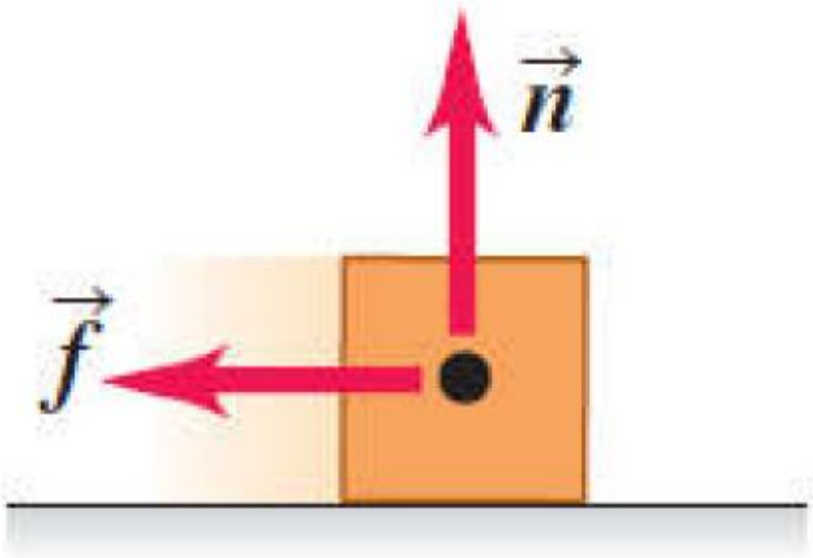
- Normal force ***n***: When an object is resting or being pushed on a surface, the surface exerts a push on it that is directed perpendicular to the surface.





Friction Force 摩擦力

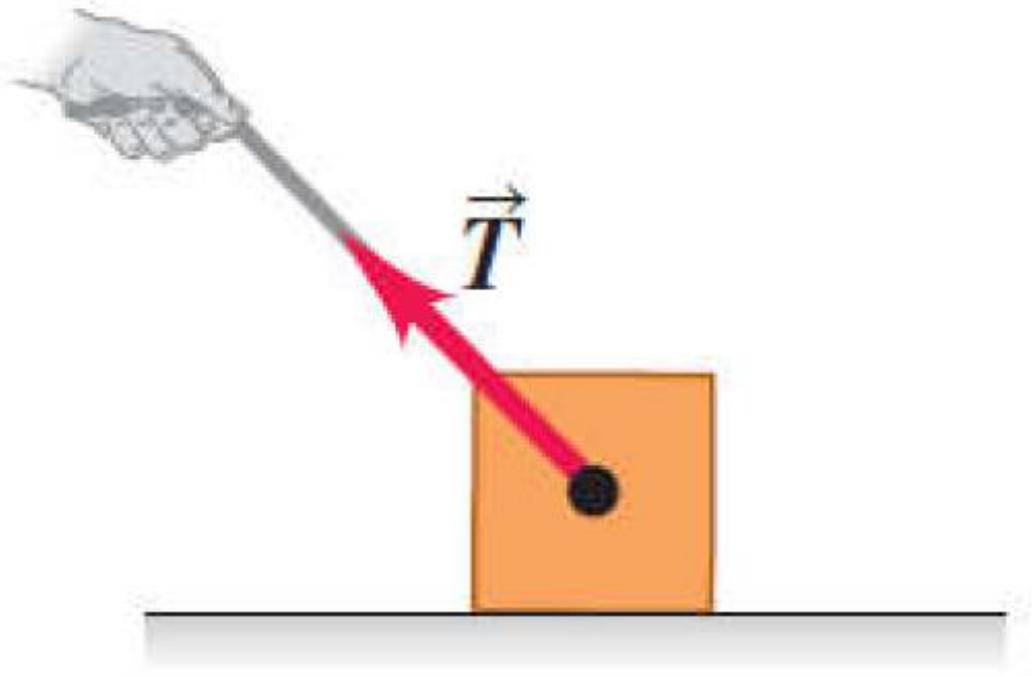
- Friction force ***f***: In addition to the normal force, a surface also exert a frictional force on the object, directed in parallel to the surface.



拉紧的

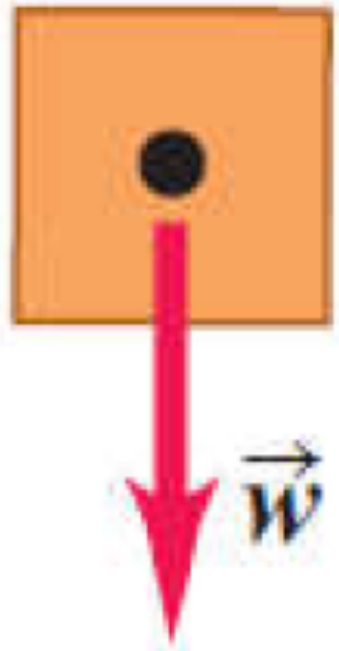
Tension Force 拉力

- Tension force ***T***: A pulling force exerted on an object by a rope, cord, etc.



Weight 重力

- Weight ***w***: The pull of gravity on an object is a long-range force (a force that acts over a distance). The gravitational force that the earth exerts on your body is your weight.



mess (kg)

$\vec{w} = mg$ (9.8)

What Are the Magnitudes of Common Forces?

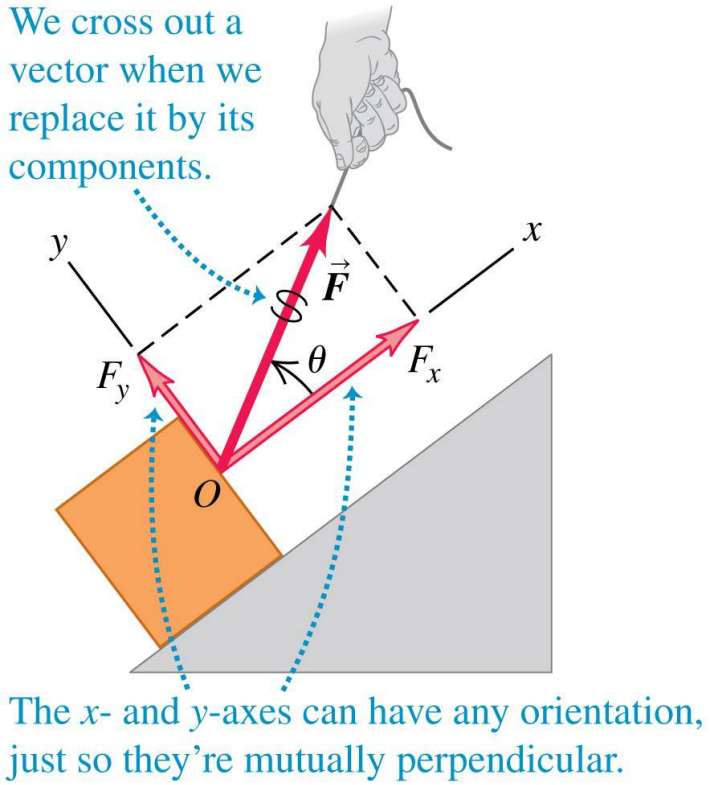
- The SI unit of the magnitude of force is the newton, abbreviated N. Some typical force magnitudes are:

Sun's gravitational force on the earth	3.5×10^{22} N
Weight of a large blue whale	1.9×10^{6} N
Maximum pulling force of a locomotive	8.9×10^{5} N
Weight of a 250-lb linebacker	1.1×10^{3} N
Weight of a medium apple	1×10^{0} N
Weight of the smallest insect eggs	2×10^{-6} N
Electric attraction between the proton and the electron in a hydrogen atom	8.2×10^{-8} N
Weight of a very small bacterium	1×10^{-18} N
Weight of a hydrogen atom	1.6×10^{-26} N
Weight of an electron	8.9×10^{-30} N
Gravitational attraction between the proton and the electron in a hydrogen atom	3.6×10^{-47} N

分解

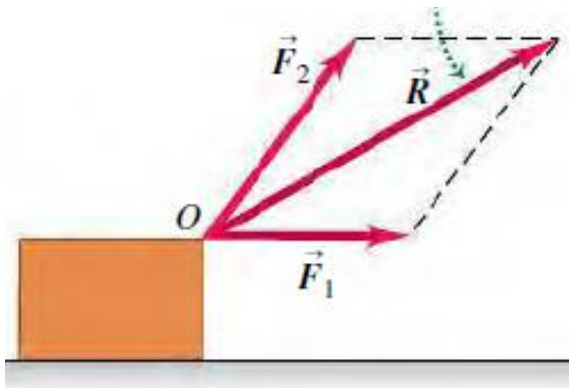
Decomposing a Force into Its Component Vectors

- Choose perpendicular x- and y-axes.
- F_x and F_y are the components of a force along these axes.
- Use trigonometry to find these force components.



Superposition of Force

- Superposition of force is the resultant of two forces acting on the same object: $\vec{R} = \vec{F}_1 + \vec{F}_2$

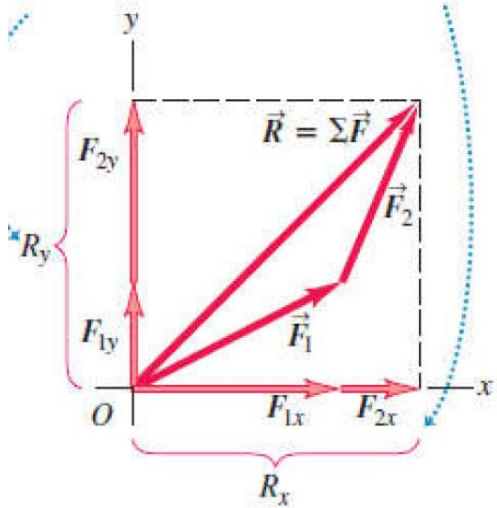


- Finding the components of the vector sum (resultant), all forces acting on a body:

The net force acting on a body ... $\vec{R} = \Sigma\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \cdots$ (4.1)

... is the vector sum, or resultant, of all individual forces acting on that body.

$$R_x = \Sigma F_x \qquad R_y = \Sigma F_y \tag{4.2}$$



- Finding the magnitude and direction:

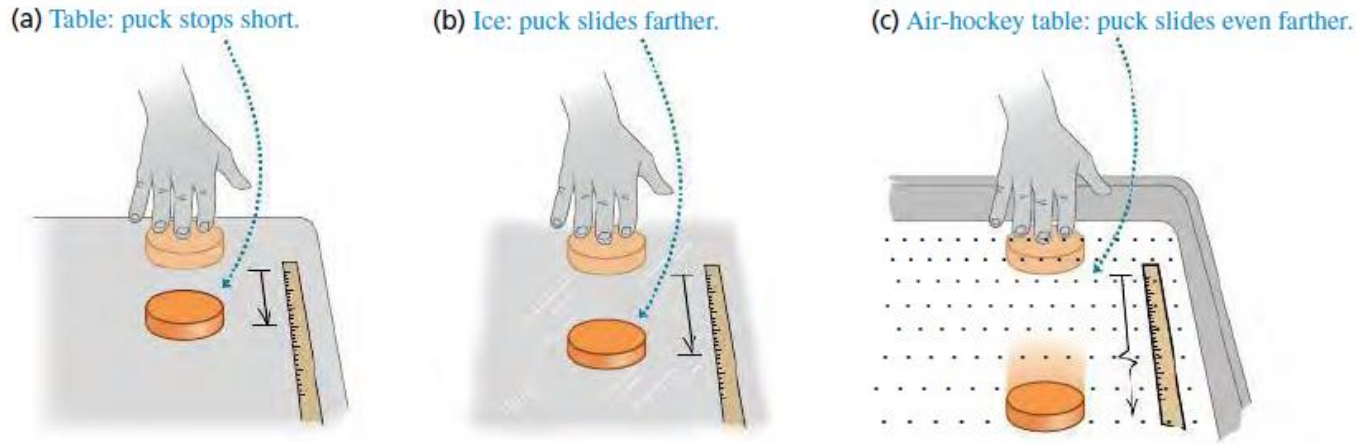
$$R = \sqrt{R_x^2 + R_y^2} \qquad \tan\theta = R_y/R_x.$$

$$R = \sqrt{R_x^2 + R_y^2 + R_z^2}$$

Newton's First Law of Motion

- Definition: A body acted on by no net force has a constant velocity (which may be zero) and zero acceleration.

Newton's first law: Net force on a body ... $\sum \vec{F} = 0$ ... must be zero if body is in equilibrium. (4.3)

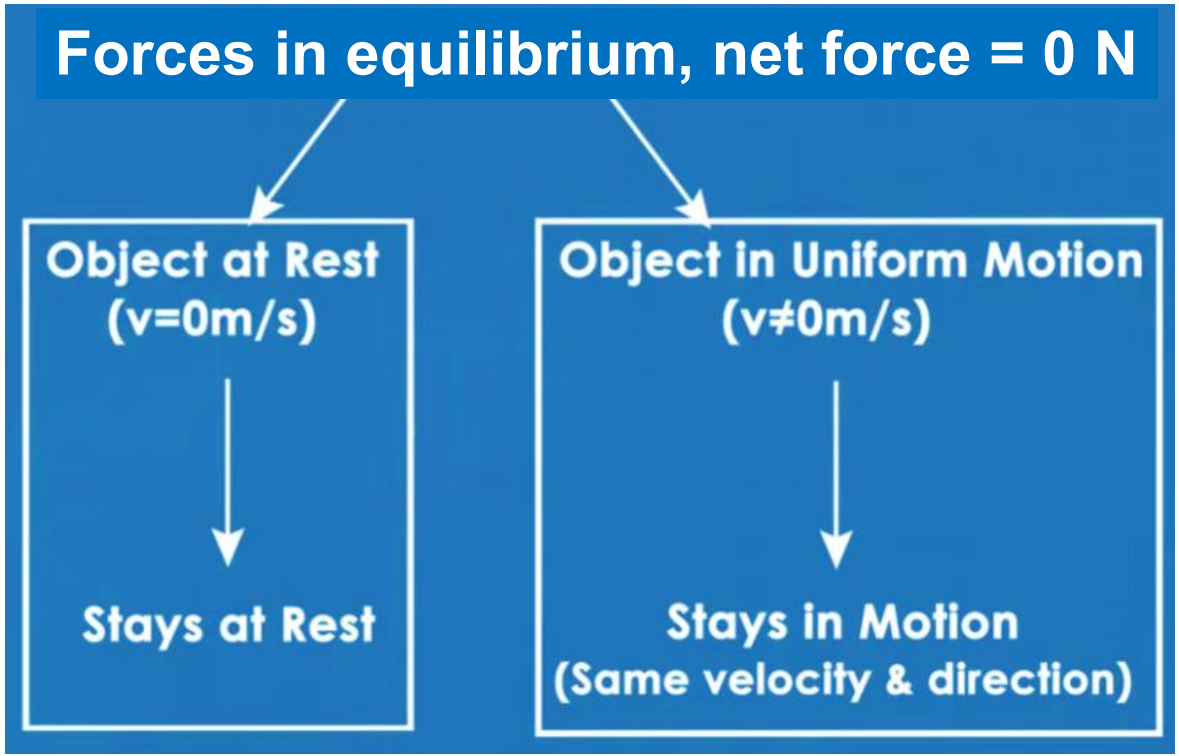


- The slicker the surface, the farther a puck slides after being given an initial velocity. On an air-hockey table (c) the friction force is practically zero, so the puck continues with almost constant velocity.

Newton's First Law of Motion | Forces and Motion | Physics | Don't Memorise - YouTube

Newton's First Law of Motion

- In another words, an object at rest stays at rest, and an object in motion stays in motion, with the same velocity and direction, unless acted upon by an external force.



Newton's First Law of Motion

- A scary situation we often see in space fiction movies: an astronaut drifts away in space because there are ZERO friction/forces.
- A single collision/push will cause them to drift away in space indefinitely.

Astronauts are tethered with a spring loaded long tether and also have a secondary tether. They are double checked to make sure every thing is OK.



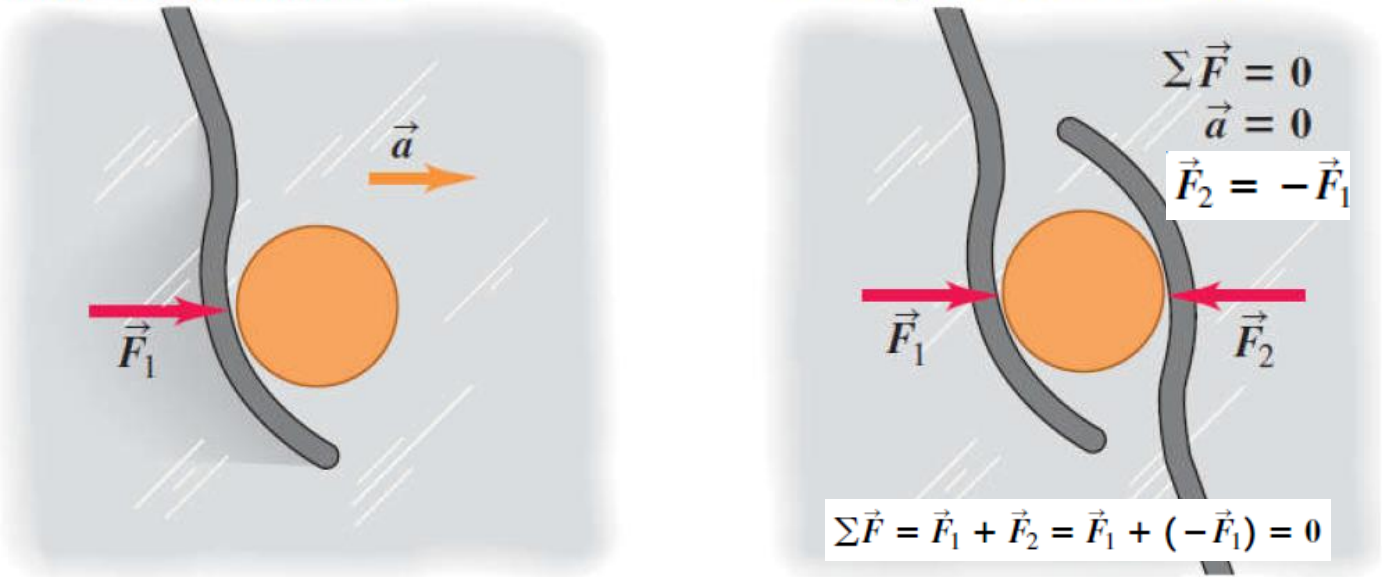
In case the tether breaks, they have a maneuvering unit called SAFER (Simplified Aid for EVA Rescue) to bring them back to airlock.

Newton's First Law of Motion

(a) A hockey puck accelerates in the direction of a net applied force F_1. (b) When the net force is zero, the acceleration is zero, and the puck is in equilibrium.

(a) A puck on a frictionless surface accelerates when acted on by a single horizontal force.

(b) An object acted on by forces whose vector sum is zero behaves as though no forces act on it.



Newton's First Law of Motion

- The downward force of gravity acting on the child and sled is balanced by an upward normal force exerted by the ground.
- The adult's foot exerts a forward force that balances the backward force of friction on the sled.
- Hence there is no net force on the child and sled, and they slide with a constant velocity.



First Law and Inertia 惯性

- The tendency of an object to resist any attempt to change its velocity is called the inertia of the object.
- Inertia is related to the mass m. The bigger the mass of an object, the bigger the inertia of the object.

m↑ inertia↑





— You can start very quickly and then run faster even than the car at the beginning !

- Mass is an inherent property of a body and it is independent of the body's surroundings and method used to measure it. It is a scalar quantity.

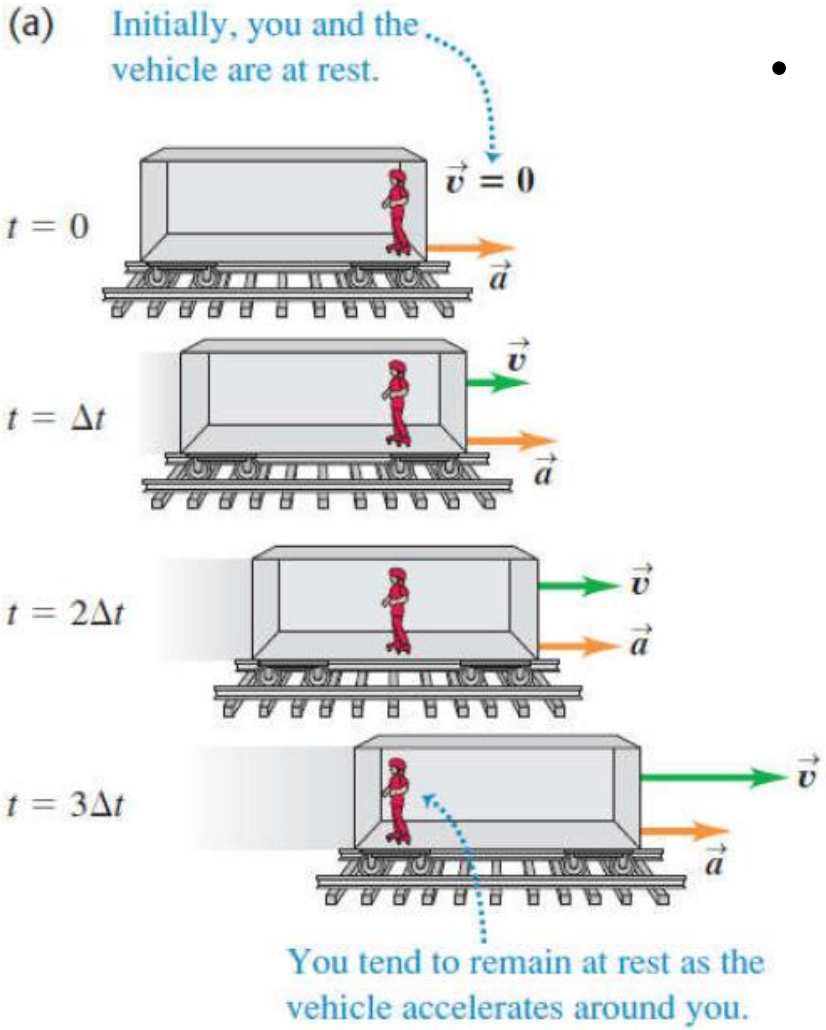
First Law and Inertia





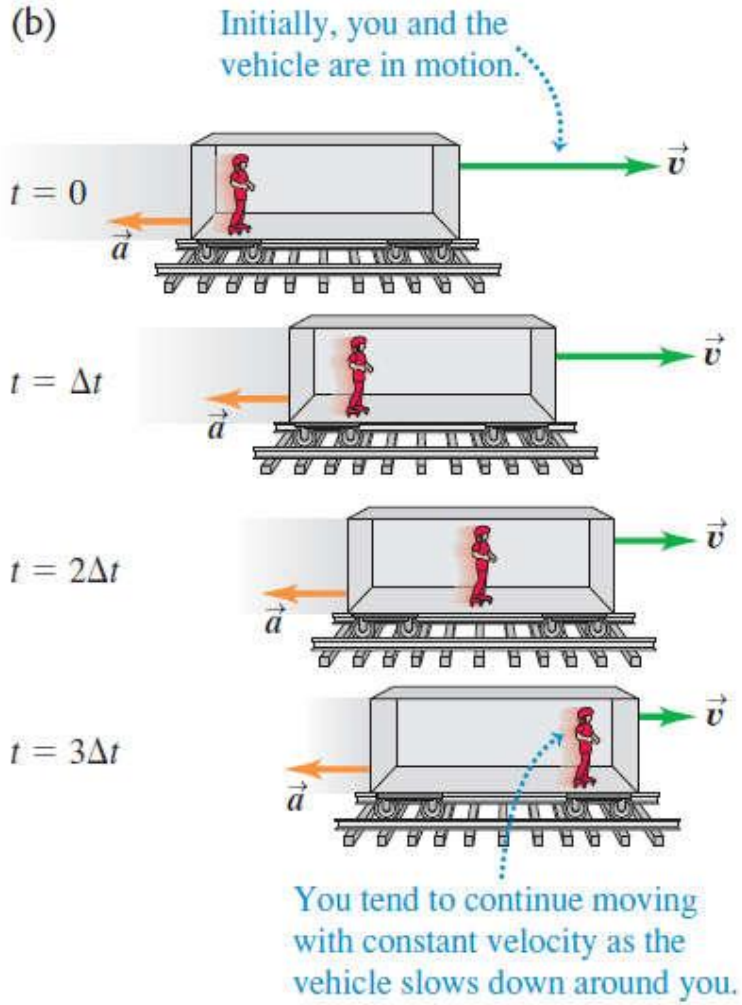


First Law and Inertia



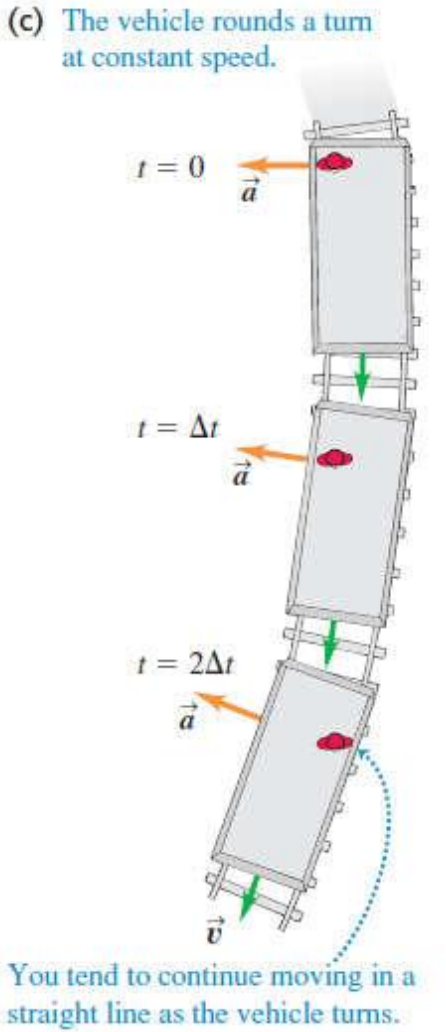
- When riding in a vehicle that's **accelerating,** there is no net force acting on you, yet your velocity changes, why the First Law is not obeyed?
  - A vehicle was initially at rest and then begins to **accelerate** to the right.
  - A passenger on ***roller skates*** (which nearly eliminate the effects of friction) has virtually **no net force acting** on her, so she tends to remain at rest relative to the inertial frame of the earth.
  - As the vehicle accelerates around her, she moves backward relative to the vehicle.

First Law and Inertia



- In the same way, a passenger in a vehicle that is slowing down tends to continue moving with constant velocity relative to the earth, and so moves forward relative to the vehicle.

First Law and Inertia



- A vehicle is also accelerating if it moves at a constant speed but is turning.
- The earth at constant speed is a straight line, relative to the vehicle.
- The passenger moves to the side of the vehicle on the outside of the turn.

Inertial Frame of Reference

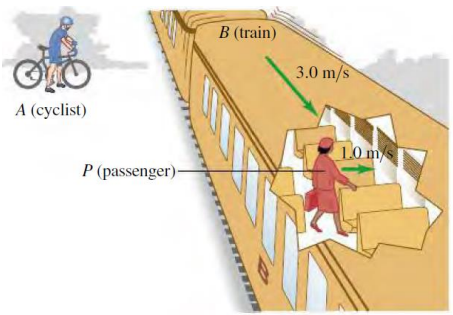
- Why the First Law is Not obeyed?
  - In each case shown in the previous slides, an observer in the vehicle's frame of reference might be tempted to conclude that there is a net force acting on the passenger, since the passenger's velocity relative to the vehicle changes in each case.
  - **This conclusion is simply wrong**; the net force on the passenger is indeed zero. The vehicle observer's mistake is in trying to apply Newton's first law in the vehicle's frame of reference, which **is not an inertial frame** and in which Newton's first law isn't valid.
- So what is an inertial frame of reference?

Inertial Frame of Reference

- A frame of reference in which Newton's first law is valid is called an inertial frame of reference, i.e., an unaccelerated frame.
- What/Who can be an inertia frame?
  - The earth/the surface of the earth is an inertia frame. Although the earth is just an approximate inertia frame of reference, normally we assume that a set of nearby points on the surface of the Earth as the inertia frame of reference.
  - Any frame that moves with a constant velocity to an inertia frame, such as moving with constant velocity to the earth/surface of the earth, is itself an inertia frame.

Inertial Frame of Reference

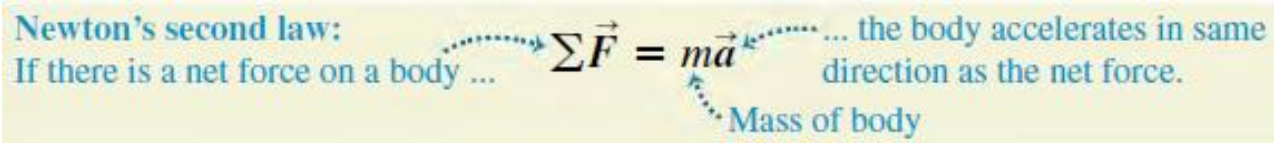
- Referring to the relative-velocity relationship in Eq (3.35).
- Suppose that P is a body that moves with constant velocity $\vec{v}_{P/A}$ with respect to an inertial frame A, by Newton's first law the net force on this body is zero.
- The velocity of P relative to another frame B is given by: $\vec{v}_{P/B} = \vec{v}_{P/A} - \vec{v}_{B/A}$
- But if the relative velocity $\vec{v}_{B/A}$ of the two frames is constant, then $\vec{v}_{P/B}$ is constant as well.
- Thus B is also an inertial frame; the velocity of P in this frame is constant, and the net force on P is zero, so Newton's first law is obeyed in B.



- Observers in frames A and B will disagree about the velocity of P, but they will agree that P has a constant velocity (zero acceleration) and has zero net force acting on it.
- Newton's First Law is sometimes called law of Inertia.

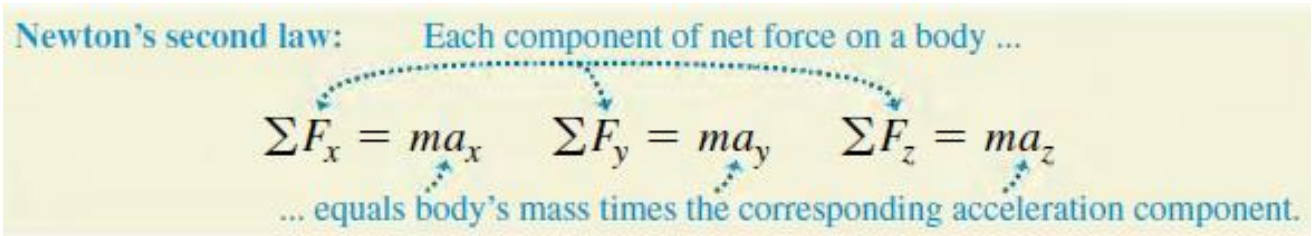
Newton's Second Law – Mass and Force

- Definition: If a net external force acts on a body, the body accelerates. The direction of acceleration is the same as the direction of the net force. The mass *m* of the body times the acceleration vector of the body equals the net force vector.



(4.6)

- Since it is a vector equation, we can separate the equation for each component of force and the corresponding component of acceleration:



(4.7)

- Eqs (4.6) and (4.7) valid only when the mass *m* is constant.
- Newton's Second Law is a fundamental law of nature, the basic relationship between force and motion.

Newton's Second Law of Motion | Physics | Don't Memorise - YouTube

Newton's Second Law – Mass and Force

- The design of high-performance motorcycles depends fundamentally on Newton's second law.
- To maximize the forward acceleration, the designer makes the motorcycle as light as possible (that is, minimizes the mass) and uses the most powerful engine possible (thus maximizing the forward force).



$$\vec{a} = \frac{\sum \vec{F}}{m}$$

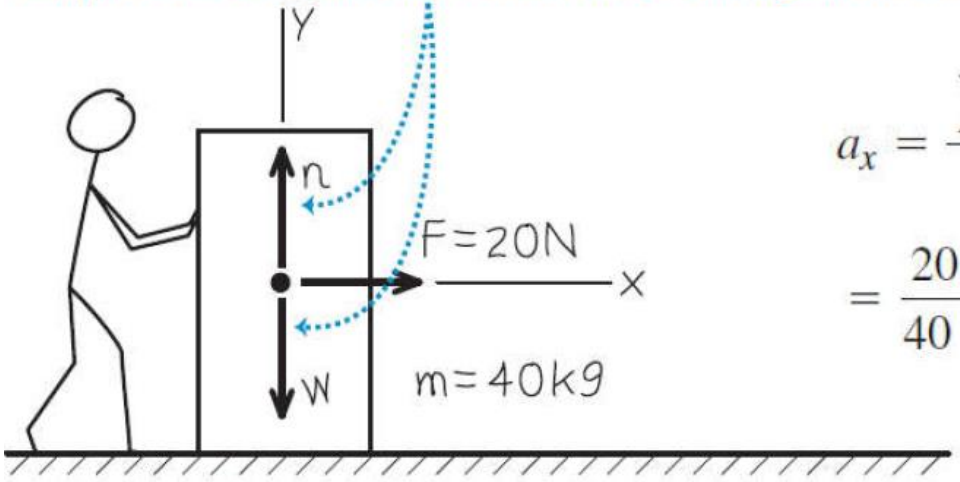
Newton's Second Law – Mass and Force

Example 4.4 Determining acceleration from force

A worker applies a constant horizontal force with magnitude 20 N to a box with mass 40 kg resting on a level floor with negligible friction. What is the acceleration of the box?

可忽略的

The box has no vertical acceleration, so the vertical components of the net force sum to zero. Nevertheless, for completeness, we show the vertical forces acting on the box.



$$a_x = \frac{\sum F_x}{m}$$

$$= \frac{20\ \text{N}}{40\ \text{kg}} = \frac{20\ \text{kg}\cdot\text{m/s}^2}{40\ \text{kg}} = 0.50\ \text{m/s}^2$$

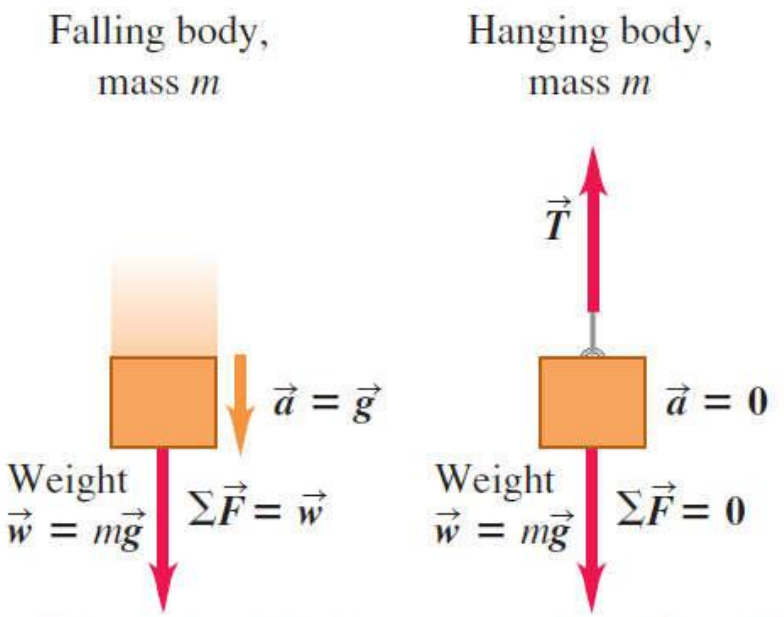
Newton's Second Law – Mass and Weight

- Mass *m* characterizes the inertial properties of a body.
- Weight ***w*** is a force that exerts on the body by the pull of the earth.
- Weight is a vector, but mass is a scalar.

Magnitude of weight of a body: $w = mg$ (4.8) — Mass of body; Magnitude of acceleration due to gravity

- Remember that *g* is the magnitude of the acceleration due to gravity, so *g* is always a positive number, by definition.
- Thus the magnitude of the weight *w* is also always positive.

$$\vec{w} = m\vec{g} \qquad (4.9)$$



- The relationship of mass to weight: $\vec{w} = m\vec{g}$.
- This relationship is the same whether a body is falling or stationary.

Newton's Second Law – Mass and Weight

Example 4.7 Mass and weight

A 2.49×10^4 N Rolls-Royce Phantom traveling in the $+x$-direction makes an emergency stop; the x-component of the net force acting on it is -1.83×10^4 N. What is its acceleration?

EXECUTE: The mass of the car is

$$m = \frac{w}{g} = \frac{2.49 \times 10^4 \text{ N}}{9.80 \text{ m/s}^2} = \frac{2.49 \times 10^4 \text{ kg} \cdot \text{m/s}^2}{9.80 \text{ m/s}^2}$$
$$= 2540 \text{ kg}$$

Then $\sum F_x = ma_x$ gives

$$a_x = \frac{\sum F_x}{m} = \frac{-1.83 \times 10^4 \text{ N}}{2540 \text{ kg}} = \frac{-1.83 \times 10^4 \text{ kg} \cdot \text{m/s}^2}{2540 \text{ kg}}$$
$$= -7.20 \text{ m/s}^2$$

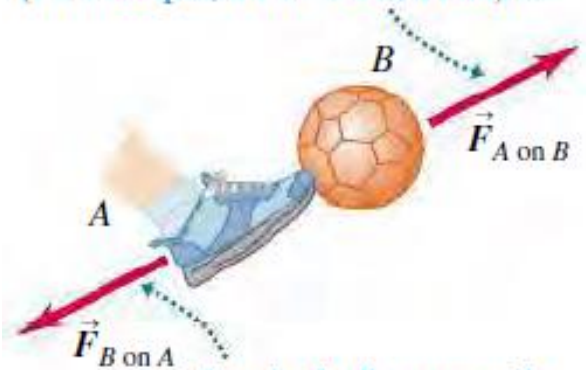
Caution: DO NOT confuse mass and weight. The SI units for mass and weight are often misused in everyday life. For example, it's incorrect to say "This box weighs 6 kg"; what this really means is that the mass of the box, probably determined indirectly by weighing, is 6 kg. Avoid this sloppy usage in your own work! In SI units, weight (a force) is measured in newtons, while mass is measured in kilograms.

Newton's Third Law

- Definition: If body A exerts a force on body B (an "action"), then body B exerts a force on body A (a "reaction"). These two forces have the same magnitude but are opposite in direction. These two forces act on different bodies.

$$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A} \qquad \text{(Newton's third law of motion)} \qquad (4.10)$$

If body A exerts force $\vec{F}_{A \text{ on } B}$ on body B (for example, a foot kicks a ball) ...



... then body B necessarily exerts force $\vec{F}_{B \text{ on } A}$ on body A (ball kicks back on foot).

The two forces have same magnitude but opposite directions: $\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$.

Newton's Third Law of Motion | Forces and Motion | Physics | Don't Memorise - YouTube

Newton’s Third Law



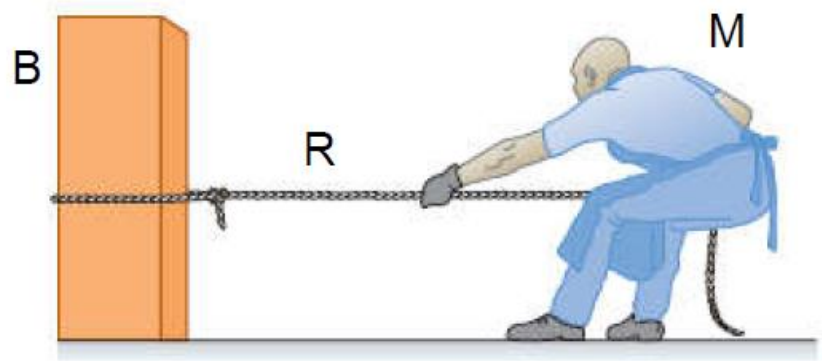
- The simple act of walking depends crucially on Newton’s third law.
- To start moving forward, you push backward on the ground with your foot.
- As a reaction, the ground pushes forward on your foot (and hence on your body as a whole) with a force of the same magnitude.
- This external force provided by the ground is what accelerates your body forward.

Newton's Third Law

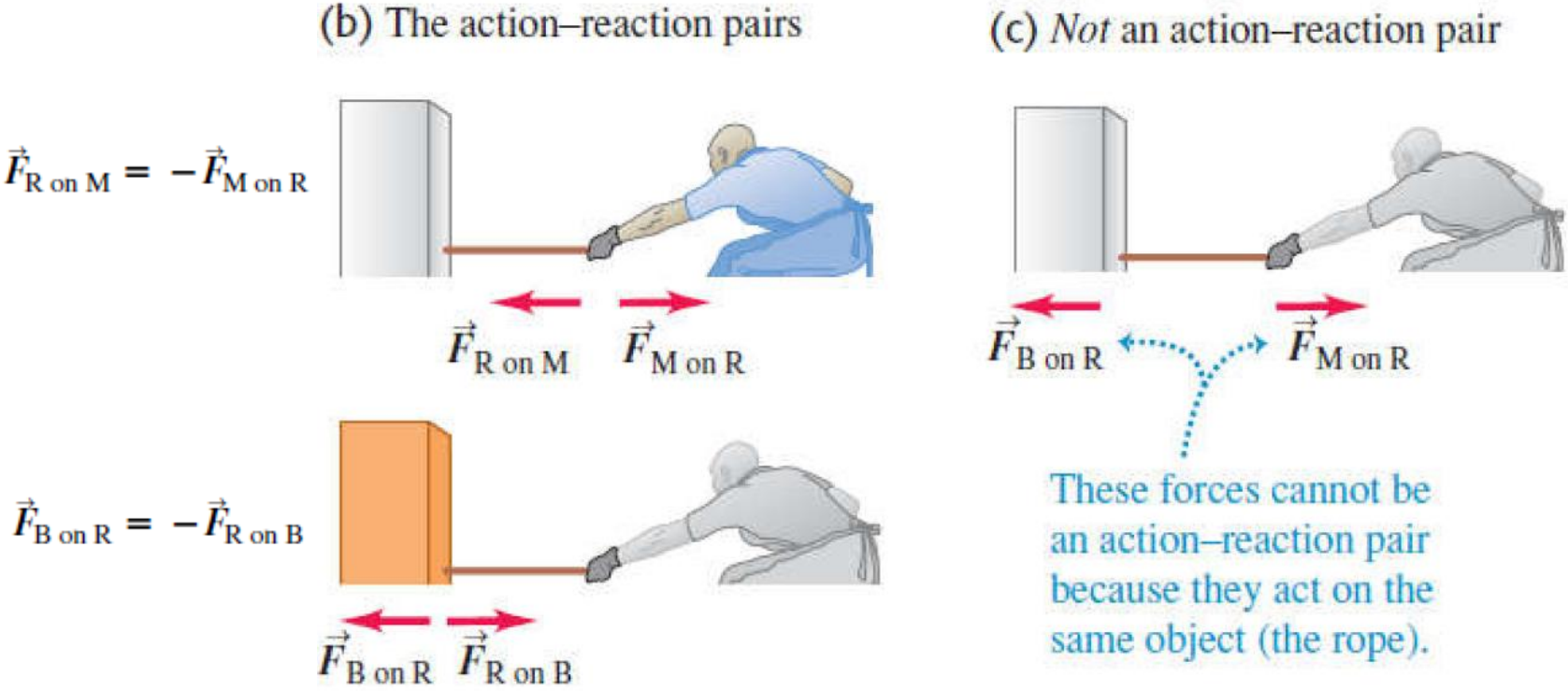
Conceptual Example 4.10 **Applying Newton's third law: Objects in motion**

A stonemason drags a marble block across a floor by pulling on a rope attached to the block (Fig. 4.27a). The block is not necessarily in equilibrium. How are the various forces related? What are the action–reaction pairs?

(a) The block, the rope, and the mason



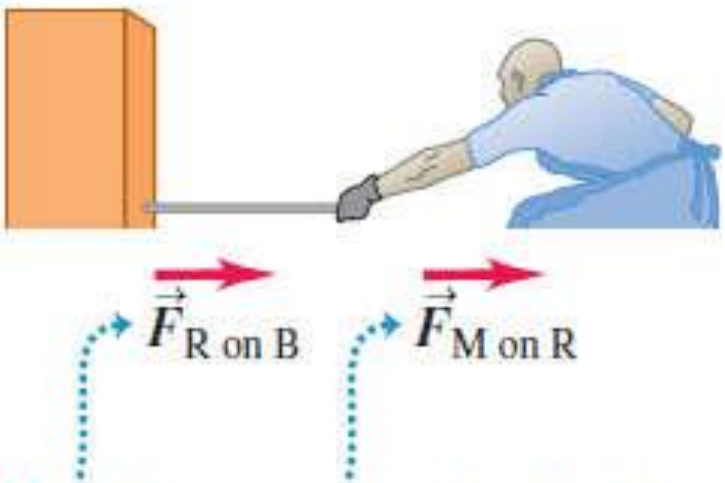
Newton's Third Law



- The forces acting on the same object cannot be action-reaction pairs.
  - The two forces in an action–reaction pair never act on the same body.

Newton's Third Law

(d) Not necessarily equal



These forces are equal only if the rope is in equilibrium (or can be treated as massless).

The forces $F_{\text{M on R}}$ and $F_{\text{B on R}}$ are not necessarily equal in magnitude. Applying Newton's second law to the rope, we get

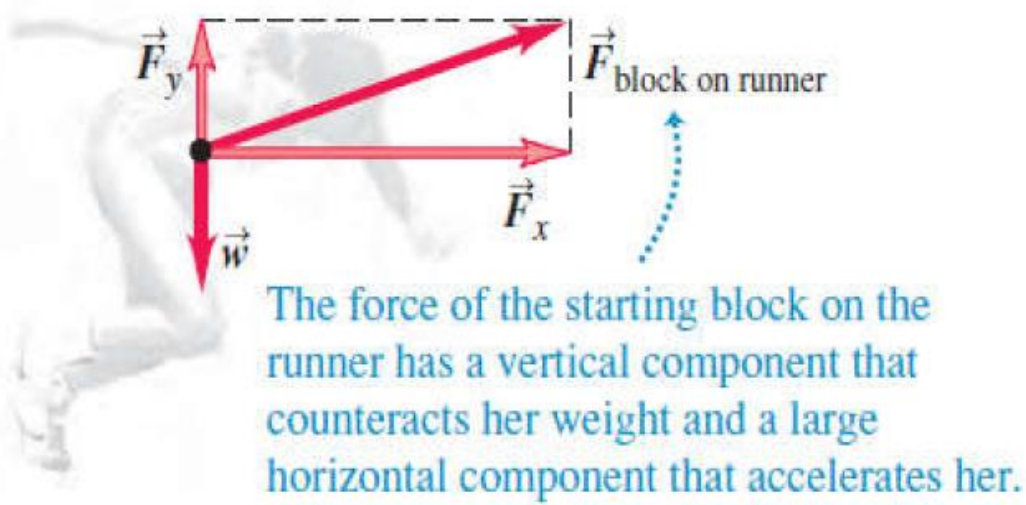
$$\sum \vec{F} = \vec{F}_{\text{M on R}} + \vec{F}_{\text{B on R}} = m_{\text{rope}}\vec{a}_{\text{rope}}$$

$$\vec{F}_{\text{B on R}} = -\vec{F}_{\text{R on B}}$$

Free Body Diagram

- Free body diagrams are essential to help identify the relevant forces.
- It is a diagram showing the chosen body by itself, “free” of its surroundings, with vectors drawn to show the magnitudes and directions of all the forces applied to the body by the various other bodies that interact with it.

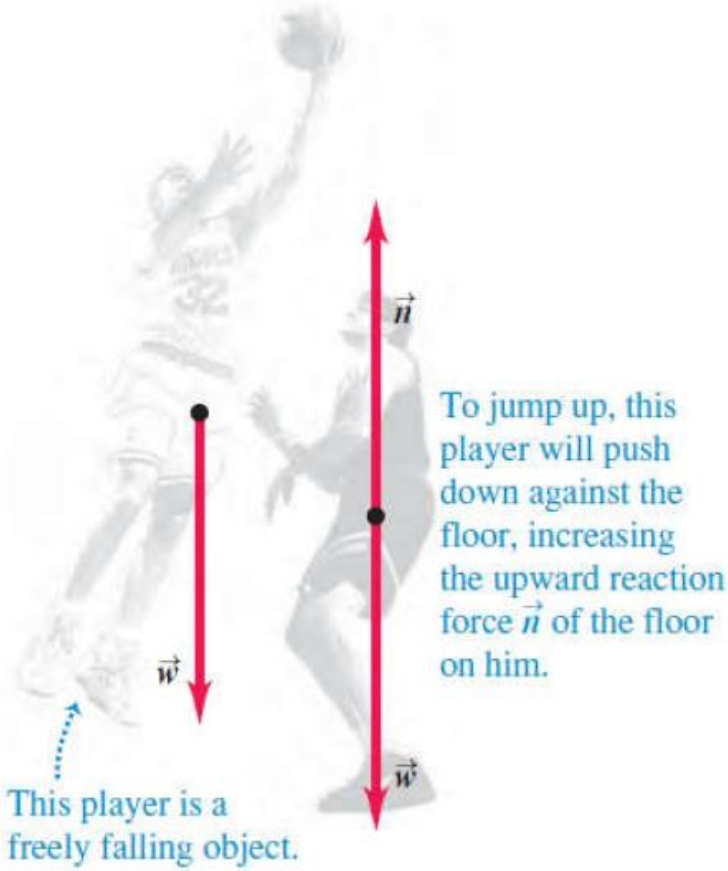




Free Body Diagram

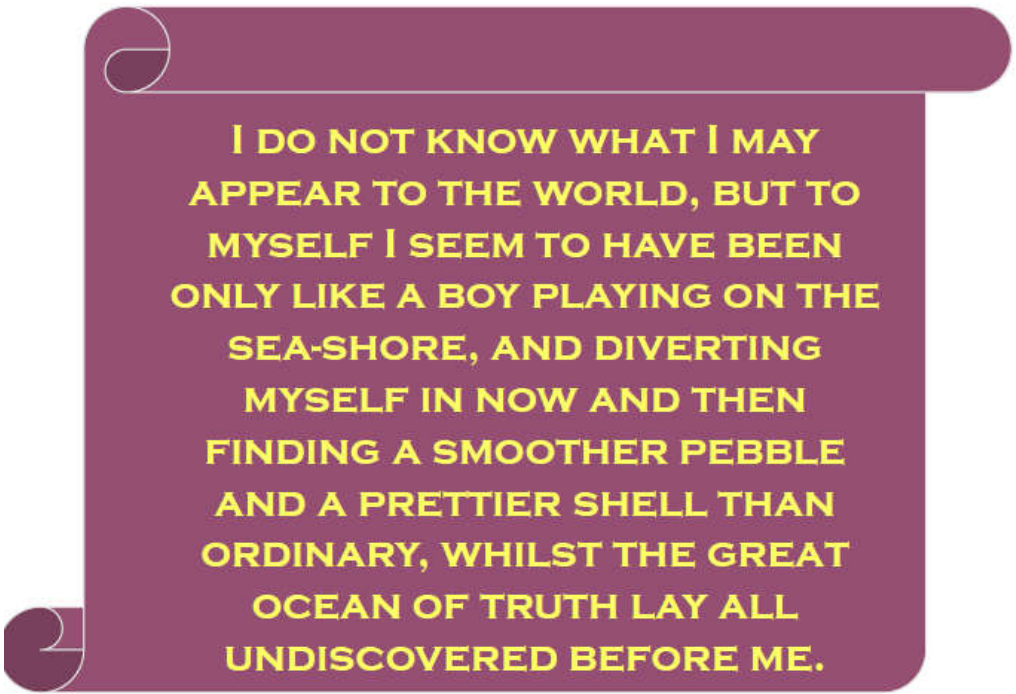
- Before we start to use the Newton's Laws to solve the question, we need to draw the Free Body Diagram of the object for the analysis about all the forces acting on the object.





Sir Isaac Newton

Sir Isaac Newton (1642 – 1727)



In memory of Sir Isaac Newton, one of the greatest physicists in human history.

Summary

- Newton’s Laws of Motions are discussed and studied.
- Textbook: Chapter 4, Newton’s Laws of Motion